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Master Thesis

Easy-to-Read (E2R) Adaptation of Figurative Language

Author: **Jelle van Lieshout**

Supervisor: **Dr. Mari Carmen Suárez-Figueroa**

Madrid, MM YYYY

Abstract

Figurative language, such as idioms and metaphors, presents a significant barrier to text accessibility and downstream Natural Language Processing tasks. While Large Language Models (LLMs) have shown promise in general language understanding, their ability to reliably handle figurative expressions in zero-shot scenarios remains under-explored compared to task-specific supervised models. This thesis investigates the effectiveness of agentic LLM pipelines for the detection, interpretation, and Easy-to-Read (E2R) adaptation of figurative language.

We propose a multi-stage agentic workflow that explicitly decomposes the task into detection, explanation, literal replacement, and self-verification. Using the SemEval-2022 Task 2 dataset for idioms and the VU Amsterdam Metaphor Corpus for metaphors, we evaluate the system’s performance against both monolithic single-prompt LLM baselines and established supervised benchmarks.

We hypothesize that an agentic decomposition significantly improves the quality of literal replacements compared to standard prompting, particularly for complex metaphors that rely on abstract source-target mappings. Furthermore, we explore the trade-offs between semantic fidelity and readability in the generated replacements. This work aims to demonstrate that structured, observable agentic reasoning enables reliable figurative language interpretation and transformation without the need for extensive task-specific training, thereby contributing to more accessible and inclusive digital content.

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2.4 Datasets

2.4.1 VU Amsterdam Metaphor Corpus (VUAMC)

The VU Amsterdam Metaphor Corpus [12] is a large, manually annotated corpus of English texts drawn from four genres: academic prose, news, fiction, and conversation. Metaphor annotation follows the MIP (Metaphor Identification Procedure) [13]: for each content word, annotators assess whether its contextual meaning contrasts with a more basic, concrete meaning from which the contextual meaning can be derived by comparison. Words meeting this criterion are labelled metaphorical.

The corpus is distributed as an XML file and contains 16,202 English sentences, each with per-token metaphoricity labels (binary: metaphorical / non-metaphorical). Crucially, the VUAMC provides *token-level* annotations rather than span- or sentence-level ones, making it well-suited for evaluating fine-grained detection but challenging for span-based models, since a single conceptual metaphor may involve multiple individually annotated tokens.

A custom Python client was developed to parse the XML, extract sentence-token structures, and derive character-offset spans from the token annotations. All 16,202 sentences are ingested into the experiment database. The VUAMC does *not* include gold literal paraphrases; replacement evaluation on this dataset therefore relies on detection metrics and human evaluation only.

2.4.2 SemEval-2022 Task 2: Multilingual Idiomaticity Detection and Sentence Embedding

SemEval-2022 Task 2 [16] is a shared task on idiom detection and interpretation in English, Portuguese, and Galician. This thesis uses the English (EN) partition of Subtask A and Subtask B:

- **Subtask A** provides sentences containing a pre-identified multi-word expression (MWE) and asks whether the MWE is used idiomatically in context (binary classification). Gold annotations include character-offset span boundaries for the MWE.
- **Subtask B** provides gold literal paraphrases of the full sentence for a subset of Subtask A examples, enabling evaluation of replacement quality.

The English partition contains 3,487 sentences after filtering and joining Tasks A and B. Gold paraphrases from Task B are stored alongside each example and used as references for BLEU and BERTScore evaluation. Unlike the VUAMC, SemEval 2022 Task 2 is evaluated at the sentence and span level, with no per-token labels.

2.4.3 Other Figurative-Language Datasets

- MAGPIE [7]: idiom interpretation robustness (qualitative / stress testing)
- TroFi [3]: verb metaphor detection (focused evaluation)
- MOH / MOH-X [9]: verb metaphor detection (secondary validation)
- MetaNet: conceptual metaphor grounding (interpretation support)

2.5 Evaluation Methodology

Standard automatic metrics are commonly applied to generation and paraphrase tasks, but their suitability for figurative language is limited. This section outlines the key metrics used in this thesis and discusses their limitations in the context of figurative-to-literal adaptation.

2.5.1 Automatic Metrics

BLEU. BLEU measures n-gram overlap between a system output and one or more reference texts [10]. It is precision-oriented and applies a brevity penalty. For figurative language replacement, BLEU is severely misaligned: literal paraphrases of idiomatic expressions intentionally avoid lexical overlap with the gold reference, and multiple semantically equivalent paraphrases may share no surface n-grams. BLEU is retained here primarily as a baseline point of comparison, but its low values should not be interpreted as evidence of poor replacement quality.

SARI. SARI decomposes quality into Add, Delete, and Keep operations relative to the input and references, and was originally designed for text simplification [17]. Figurative-to-literal transformation closely resembles simplification (replacing opaque expressions with transparent ones), making SARI conceptually well-motivated. It is nevertheless not used in this thesis: its reference dependence makes it inapplicable to the metaphor track (the VUAMC has no gold paraphrases), and on the idiom track it would be redundant with the human evaluation, which is the load-bearing instrument for replacement quality.

BERTScore. BERTScore computes contextual token-embedding similarity between generated and reference texts using a pre-trained language model (RoBERTa in this work), enabling soft alignment rather than exact string matching [18]. This makes it more robust to paraphrase variation than BLEU. However, high BERTScore does not guarantee correct interpretation: a plausible but semantically incorrect paraphrase may still achieve a high score. BERTScore is used as the primary automatic metric for replacement evaluation.

2.5.2 Human Evaluation

Automatic metrics alone are insufficient for assessing figurative-to-literal replacement quality. Recent work in sentence simplification has shown that most automatic metrics correlate only spuriously with human judgement on this class of task [2, 11], motivating a primary reliance on human evaluation. The protocol adopted is *Direct Assessment* [5, 6], applied to the three-axis rubric (grammaticality, meaning preservation, simplicity) established for sentence simplification by Alva-Manchego et al. [1]. Human evaluation is reserved for the open-ended task of replacement, where no single ground truth exists; detection has gold annotations and is evaluated only with the automatic metrics above. The replacement evaluation covers idioms only (SemEval Task B); metaphors are evaluated through detection metrics on detect-then-replace runs (Section 7). The full survey methodology — continuous-scale design, standardisation and aggregation, quality

control, and three deliberate departures from the original DA specification — is the subject of Chapter 4; the survey instantiation (system conditions, source-sentence pool, respondent recruitment) is part of the RQ3 experimental setup (Section 7.5).

2.5.3 Multi-Layer Evaluation Framework

The evaluation protocol is structured as four complementary layers, inspired by the framework proposed in [14]:

1. **Detection accuracy:** span-level precision, recall, and F1 (Section 3.4);
2. **Semantic similarity:** BERTScore and BLEU (supporting/automatic);
3. **Interpretive correctness:** human evaluation of meaning fidelity (primary);
4. **Conceptual grounding:** correctness of source–target metaphor mappings and idiom sense disambiguation (planned).

2.6 Synthesis: The Gap This Thesis Fills

3 System Design & Methodology

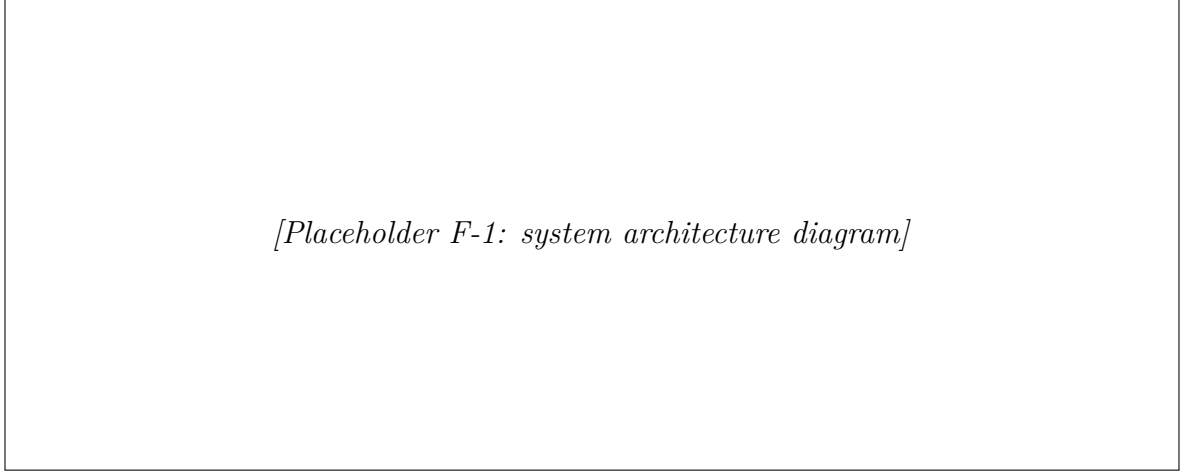


Figure 1: System architecture: FastAPI experiment runner, LangGraph workflows, Couchbase persistence (datasets, runs, predictions, evaluations), vLLM open-weights serving on the UPM A100 node, LangSmith tracing, and the React inspection frontend.

3.1 Problem Formulation

The core task addressed by this thesis is figurative language adaptation: given a raw English sentence that may contain figurative expressions (idioms or metaphors), produce a literal, meaning-preserving restatement alongside explicit detection annotations. More formally, the system accepts a sentence s and outputs:

- A set of detected figurative spans $\{(e_i, \text{start}_i, \text{end}_i)\}$, where each e_i is a verbatim copy of an expression from s with character-offset boundaries;
- A per-token binary label sequence $\mathbf{l} \in \{0, 1\}^{|\mathcal{T}(s)|}$ over the whitespace-tokenised form of s , derived from the detected spans;
- A brief linguistic explanation of the figurative usage;
- Optionally, a literal paraphrase $\hat{s}$ of the full sentence, with all figurative expressions replaced by meaning-preserving literal alternatives.

Two task variants are supported: *detection* (spans, labels, and explanation only) and *detect-then-replace* (additionally producing $\hat{s}$). Both are evaluated over two phenomena independently: *metaphors*, benchmarked against the VU Amsterdam Metaphor Corpus [12], and *idioms*, benchmarked against SemEval 2022 Task 2 [16]. A notable challenge is that some metaphors encode meaning that resists full literalisation without loss; replacement decisions therefore require nuanced linguistic understanding beyond surface paraphrase.

3.2 Agentic Pipeline

The system is implemented as four LangGraph workflows, one per (phenomenon, task) combination:

- `metaphor_detection_graph` — detects metaphorical words or phrases, following VUAMC MIP (Metaphor Identification Procedure) guidelines;
- `idiom_detection_graph` — detects multi-word idiomatic expressions, following SemEval 2022 Task 2 annotation conventions;
- `metaphor_detect_then_replace_graph` — detects metaphors and produces a literal sentence paraphrase;
- `idiom_detect_then_replace_graph` — detects idioms and produces a literal sentence paraphrase.

Each workflow issues a single structured LLM call whose output must conform to the `TaskOutput` Pydantic schema:

- `detection.is_figurative` (boolean): whether the sentence contains any figurative expression;
- `detection.figurative_expressions` (list of strings): verbatim copies of each detected expression, preserving casing and punctuation as they appear in the input text;
- `detection.explanation` (string): brief linguistic reasoning for the detection decision;
- `replacement.literal_paraphrase` (optional string): the complete sentence rewritten with all figurative expressions replaced.

Structured output is enforced via LangChain’s `with_structured_output()` interface, which uses JSON-schema-constrained decoding or function-calling depending on the model’s capabilities. Outputs violating the schema trigger automatic retries.

Two deterministic post-processing steps follow the LLM call:

1. **Span extraction:** each verbatim expression in `figurative_expressions` is located in the input sentence by exact string search, with a case-insensitive fallback. Matched positions are converted to `Span(start, end)` objects with inclusive-start, exclusive-end character offsets. Overlapping spans are merged; all spans are clipped to the sentence boundary.
2. **Token label derivation:** a binary label is assigned to each whitespace-delimited token by testing whether any detected span overlaps the token’s character range. This produces a label sequence directly comparable to VUAMC’s token-level metaphor annotations.

3.3 Infrastructure

The system is built on the following components:

- **Backend:** FastAPI (Python 3.13) on port 3030. Experiment runs are launched asynchronously as background tasks, enabling concurrent execution.
- **LLM access:** a provider-dispatched client wrapping LangChain’s `ChatOpenAI`. Model identifiers are routed at runtime: those prefixed with `vllm:` are served by the open-weights cluster (see below), and unprefixed identifiers pass through to OpenRouter for development-time smoke-testing. All experimental results reported in this thesis are produced by the open-weights cluster; OpenRouter is retained only as a development convenience for system-shakedown runs.

- **Open-weights serving:** vLLM (continuous batching, PagedAttention) on a UPM A100 GPU node, exposed to the experiment runner via a basic-auth-protected ngrok HTTPS tunnel. Two open-weights models are evaluated: `meta-llama/Meta-Llama-3.1-8B-Instruct` (FP16, deliverable system) and `casperhansen/llama-3.3-70b-instruct-awq` (AWQ-quantised, scale-comparator). The same OpenAI-compatible client code path serves both, so the only difference between conditions is the model identifier on the run document.
- **Persistence:** Couchbase 7.6 document store with four collections: `datasets` (gold-annotated examples; 16,202 VU Amsterdam + 3,487 SemEval = 19,689 total), `runs` (experiment metadata and status), `predictions` (per-example LLM outputs), and `evaluations` (computed metrics). Document keys follow a namespaced convention (e.g. `prediction::{\run_id}::{example_id}`).
- **Prompt versioning:** each run records the full prompt text and its SHA-256 hash, enabling exact reproducibility and detection of prompt drift between runs.
- **Observability:** LangSmith tracing is configured for all LLM invocations, providing post-hoc inspection of model inputs, outputs, token usage, and latency per step. Trace analysis is discussed qualitatively in Section 8.3.
- **Frontend:** a React/TypeScript single-page application with TanStack Query for data fetching, featuring run dashboards, example-level span visualisation with colour-coded gold/predicted overlays, literal replacement diff views, and multi-run metric comparison charts.

3.4 Evaluation Framework

Five evaluation metrics are computed automatically for each completed run:

- **Token F1:** binary precision, recall, and F1 aggregated across all token labels within a run, computed using scikit-learn’s `precision_recall_fscore_support`. Applicable only when gold token-level annotations are available (VU Amsterdam).
- **Span F1:** macro-averaged precision, recall, and F1 over detected character-offset spans. A predicted span is a true positive if it overlaps any unmatched gold span with Intersection-over-Union (IoU) ≥ 0.5 , enforcing one-to-one matching.
- **Sentence F1:** binary classification F1, where a sentence is labelled figurative if and only if at least one span is detected, for both gold and predicted.
- **BLEU:** sentence-level BLEU with NLTK smoothing (method 1) against the gold paraphrase from SemEval Task B. Applicable only to detect-then-replace runs on SemEval.
- **BERTScore** [18]: contextual embedding similarity (RoBERTa model) between predicted and gold replacement sentences, returning precision, recall, and F1. Applicable only to detect-then-replace runs on SemEval.

All metrics are persisted as `EvaluationData` Couchbase documents and are re-computable on demand via the `POST /runs/{run_id}/evaluate` endpoint. Section 2.5 discusses the appropriateness of these metrics for figurative language evaluation in more detail.

3.5 Baselines

- **Monolithic single-prompt:** a one-paragraph stub prompt that requests a literal rewrite of the input sentence with free-text output. No detection step, no explicit reasoning step, no structured output schema. This is the RQ2 control condition (Section 6) and one of the three human-evaluation survey conditions (Section 7.5).
- **Detection-only comparison:** detect-then-replace runs are compared against detection-only runs on the same examples to quantify the effect of the replacement step on detection quality (see Sections 5 and 7).

[Placeholder F-2: monolithic vs. single-call structured output vs. 3-step pipeline]

Figure 2: The three system architectures compared in this thesis. The monolithic baseline produces only a free-text rewrite; the single-call structured-output workflow commits to detected spans alongside the paraphrase; the 3-step agentic pipeline separates detection, explanation, and transformation into sequential calls with inspectable intermediate state.

4 Methodology — Human Evaluation Survey

The survey covers idiom replacement only: the VUAMC provides no gold paraphrases, and the rubric triad below is well-defined for the figurative-to-literal mapping of idioms but breaks down for conventional metaphors that may require no rewrite at all (Section 7 develops this scoping decision).

This chapter specifies the human-evaluation protocol used to assess figurative-to-literal replacement quality for RQ3 (Section 7). The replacement task lacks a single ground truth: for an idiom such as *smoking gun*, valid literal paraphrases include *decisive evidence*, *conclusive proof*, or longer rewrites that preserve sentence meaning, none of which need share surface form with each other or with a gold reference. As discussed in Section 2.5, automatic metrics in this regime correlate only spuriously with human judgement [2, 11], so human evaluation is the primary signal for replacement.

The protocol adopted here is *Direct Assessment* (DA), the WMT-standard evaluation method introduced for segment-level machine translation by Graham et al. [5] and extended to document-level reliability by Graham et al. [6]. The rubric triad layered on top of DA — grammaticality, meaning preservation, and simplicity — follows the sentence-simplification convention established by Alva-Manchego et al. [1]. Section 4.1 describes the protocol; Section 4.2 documents three deliberate departures from the original Graham specification, each justified by a difference between the MT setting and the figurative-language replacement setting; Section 4.3 situates the choice on the resource frontier described by [6]. The instantiation of the protocol — specific systems compared, source-sentence pool, item counts, and respondent recruitment — is described in the experimental setup of RQ3 (Section 7.5).

4.1 Direct Assessment Protocol

4.1.1 Continuous-scale Rating

For each evaluated item, the annotator rates the system’s literal paraphrase on three independent 0–100 continuous sliders, one per rubric dimension (defined below). The continuous scale is the foundational design choice of the protocol: as Graham et al. [5] demonstrate, replacing a discrete (e.g., 1–5) Likert-style scale with a 0–100 continuous slider unlocks two assessor-intrinsic quality-control techniques that Likert scales cannot support, raises intra-annotator κ by up to +0.101, and raises inter-annotator κ by up to +0.144 when combined with per-annotator standardisation. Crowd-sourced annotators on platforms such as Mechanical Turk reach acceptable agreement under this design without expert review.

4.1.2 Rubric Triad

The three dimensions are rated independently:

- **Grammaticality** — is the output well-formed English?
- **Meaning preservation** — does it convey the meaning of the original sentence?
- **Simplicity** — is the output more accessible than the figurative source?

Annotators may optionally provide an alternative paraphrase, retained as qualitative data for the discussion section. Annotators are blind to the system-of-origin; per-condition identifiers live only in analysis-side metadata.

4.1.3 Standardisation and Aggregation

Following Graham et al. [5], raw scores are converted to z-scores per annotator (using each annotator’s own mean and standard deviation across all rated items) before aggregation. This corrects for individual scoring tendencies — “score-low” versus “score-high” annotators rating the same outputs — and is the property that makes the continuous scale recover meaningful inter-annotator agreement where Likert-style scales do not.

Aggregation follows the micro-then-macro pattern of [6]: the mean standardised score is computed per item across its repeat ratings, then per condition across items. Per-item scores are reported as standardised mean $\pm$ 95% confidence interval, with a composite of the three rubric dimensions as a summary statistic. Confidence intervals use the Student-*t* distribution over the standardised ratings; given the small per-item repeat counts (Section 4.2.2), they are reported as indicative rather than inferential.

4.1.4 Quality Control

Two assessor-intrinsic checks are interleaved into each respondent’s set:

- **Bad-reference items** — deliberately degraded versions of system outputs, constructed by lexical substitution or noun-phrase swap that breaks meaning preservation while keeping the sentence superficially fluent. The annotator should rate the bad reference lower than the corresponding genuine system output on the targeted rubric dimension.
- **Repeat-pair items** — duplicate items presented twice in the same respondent’s set, separated by other items. The absolute score difference quantifies intra-annotator consistency.

Both techniques are introduced in [5]; the canonical crowd-sourcing implementation (using bad references inside larger HITs, scored by a per-worker difference-of-means significance test, and retaining only workers with $p < 0.05$) is described in [6]. The thesis uses fixed-threshold filters in place of the per-worker significance test, for reasons documented in Section 4.2.3.

4.2 Departures from the Original Protocol

The DA protocol as described above follows Graham et al. [5, 6] closely. Three design decisions diverge from the original specification, each motivated by a feature of the figurative-language replacement task that does not hold in the machine-translation setting Graham analyses.

4.2.1 Rubric Triad versus Fluency Omission

Graham et al. [6], Section 3.3, argue that DA fluency assessment can be safely dropped when resources are constrained: an earlier study [4] found no evidence of reference bias in DA adequacy, removing the original methodological motivation for a separate fluency dimension; the freed budget is better spent expanding the test set on the adequacy dimension.

The thesis retains a third dimension — *grammaticality* — in deliberate tension with this argument, on three grounds. First, the task is text simplification, not machine translation: the simplification literature [1] converged independently on a three-axis rubric (grammaticality, meaning preservation, simplicity) because the additional dimension carries information that adequacy alone does not capture. Second, small open-weights LLMs at the scale used here (Llama-3.1-8B and Llama-3.3-70B-AWQ; see Section 7.5) produce qualitatively different error modes from professional MT systems: agreement and tense errors, dropped articles, and clipped continuations are visible in pilot outputs and would be invisible to a meaning-only rubric. Third, the budget freed by dropping the third axis would translate to perhaps four to five

additional items in our 81-item, 22-respondent design — a marginal reliability gain that does not compensate for losing a diagnostic axis.

4.2.2 Reliability Bound and Exploratory Reporting

Graham et al. [6] establish, by self-replication on Mechanical Turk over the WMT-16 EN–ES document-level QE test set, that DA scores reach $r = 0.901$ between two independent data-collection runs at ≥ 27 repeat assessments per document (averaging ≈ 107 segment-level judgements per document). The corresponding segment-level threshold from [5] is ≥ 15 repeat assessments. These thresholds define the reliability frontier above which DA can be used as a substitute for an expert gold standard.

The thesis survey ships with a target of ≥ 3 ratings per item, well below this frontier. This is a deliberate choice driven by the scale of the deliverable: 81 items at 20–25 respondents (16 rated items + 4 quality-control items per respondent) is at the upper end of what a master’s-thesis recruitment effort can sustain through the supervisor’s and the author’s networks, and tripling the respondent count to approach Graham’s threshold is not feasible. The methodological consequence is that per-condition scores are not used to make reliability-bounded claims about absolute system quality. Instead, the survey is treated as exploratory: results are reported per condition with the standardised means defined above, and pairwise condition comparisons, paired by source sentence, are tested with the paired t -test and the Wilcoxon signed-rank test on the per-source mean standardised scores. The survey supports the qualitative claims of RQ2 (decomposition vs. monolithic) and RQ1 (within-family scale comparison) at a directional level; it does not establish absolute quality estimates.

4.2.3 Quality-control Filter Design

Graham et al. [6] embed bad-reference items inside 100-translation HITs and apply a per-worker difference-of-means significance test, retaining only workers with $p < 0.05$. The test has good power at the original HIT size: with ~ 100 items per worker including a balanced bad-reference contingent, a worker who fails to discriminate degraded from genuine outputs is reliably flagged.

The thesis design has 20 items per respondent (16 rated + 4 QC), of which only two are bad references and two are repeat pairs. With $n = 2$ per QC type, the per-respondent significance test has effectively no power: a respondent could rate both bad references higher than their corresponding genuine outputs and still fail to reach $p < 0.05$ under the t -test. Fixed-threshold filters are used instead, applied in three rules. (a) *Completion*: respondents who fully rate fewer than 10 of their 20 items are excluded. (b) *Repeat-pair consistency*: respondents whose mean absolute first-versus-second-showing difference across their repeat pairs exceeds 25 points (on the 0–100 scale) are excluded. (c) *Bad-reference discrimination*: each bad-reference’s *meaning preservation* score is compared against the respondent’s own mean meaning score across their genuine items; a respondent both of whose bad references score at or above that mean is flagged, and headline results are reported with and without flagged respondents. The bad-reference rule targets the meaning dimension because the degradations are constructed to break meaning preservation specifically while remaining superficially fluent; comparing against the respondent’s *own* mean rather than a global threshold preserves the assessor-intrinsic character of the original technique under per-annotator score variation. Threshold values are fixed in advance and reported with the survey; they are not tuned post-hoc against the rated data.

4.3 Resource Frontier

Graham et al. [6] report that document-level DA on the WMT-16 test set cost USD 109 per data-collection run on Mechanical Turk, against a USD 1,422–2,728 estimate from a professional post-editor for the same test set — a 10–20 $\times$ cost ratio in favour of DA. The thesis survey uses voluntary native-English-speaker respondents recruited through Mari Carmen’s network rather than paid crowd workers, but the same ratio operates as the conceptual budget: a post-editing-based gold standard is not feasible at master’s-thesis scale, and DA is the protocol that makes the human evaluation tractable at all.

5 RQ1: Detection & Interpretation

To what extent can a zero-/one-shot agentic LLM pipeline accurately detect and interpret idiomatic and metaphorical expressions in context?

5.1 Experimental Setup

All detection experiments use the following configuration:

- **Models:** two open-weights LLMs served via vLLM on the UPM A100 cluster — `meta-llama/Meta-Llama-3.1-8B-Instruct` (the deliverable system) and `casperhansen/llama-3.3-70b-instruct-awq` (a same-family scale comparator). Both are accessed through the same OpenAI-compatible interface via the experiment runner’s `vllm: dispatch` prefix.
- **Temperature:** 0 (greedy decoding, for reproducibility);
- **Prompt version:** v1 (zero-shot, no few-shot examples);
- **Subset size:** 200 examples per dataset, sampled from the beginning of each ingested collection.

For metaphor detection, the prompt instructs the model to follow the MIP procedure: identify each word whose contextual meaning contrasts with its most basic, concrete meaning, and return those words verbatim as `figurative_expressions`. For idiom detection, the prompt instructs the model to identify multi-word expressions whose combined meaning cannot be derived compositionally.

In both cases, the LLM output is post-processed to extract character-offset spans and derive per-token binary labels (Section 3.4). These are compared against gold annotations from the respective datasets.

5.2 Results

Table 1 reports detection-only results on both Llama-3.1-8B-Instruct and Llama-3.3-70B-Instruct-AWQ, prompt v1, $T = 0$, on the same deterministic 200-example slice (ordering by `example_id`; runs 1afd4cf1, cfe4f798, 26dc7cd2, be7dd99f). The within-family scale dimension and the per-dataset detection profile are read off the same table.

Dataset	Model	F1-sent	F1-tok	P-tok	R-tok	F1-span	Δ at scale
VU Amsterdam	8B-Instruct	0.430	0.176	0.252	0.135	0.049	—
VU Amsterdam	70B-Instruct-AWQ	0.635	0.262	0.429	0.188	0.125	+20.5pp F1-sent
SemEval 2022	8B-Instruct	0.255	—	—	—	0.141	—
SemEval 2022	70B-Instruct-AWQ	0.435	—	—	—	0.225	+18.0pp F1-sent

Table 1: Zero-shot detection results across model scales (200 examples per dataset, $T = 0$, prompt v1, deterministic example ordering). Token-level metrics are not reported for SemEval, whose gold annotations are span- and sentence-level only. Within-family scaling from 8B to 70B-AWQ yields a consistent +18–20 percentage-point lift in sentence F1 and +8pp in span F1, with token-level precision driving most of the metaphor improvement (P-tok 0.252 $\rightarrow$ 0.429).

5.3 Analysis

Sentence-level F1 is higher on metaphors than on idioms in detection-only mode at both scales. VU metaphor sentence F1 is 0.430 (8B) and 0.635 (70B); SemEval idiom sentence F1 is 0.255 (8B) and 0.435 (70B). This inverts the conventional expectation under H3 (idioms easier than metaphors) and reflects an artefact of the detection-only prompt rather than a property of the phenomena: on idioms the prompt is under-specified for the model (especially at 8B, which often returns no spans at all), whereas on metaphors the model produces broad token-level flags. Span-level F1 tells a more consistent story: at 8B, 0.141 (idiom) versus 0.049 (metaphor), a $2.9\times$ ratio in idioms’ favour; at 70B, 0.225 (idiom) versus 0.125 (metaphor), $1.8\times$. The right metric for idiom-vs-metaphor difficulty therefore depends on whether the system is being asked to flag the sentence or to localise the expression.

Span F1 is much lower than sentence and token F1 for metaphors. The $\text{IoU} \geq 0.5$ threshold is sensitive to the mismatch between character offsets derived from LLM verbatim outputs and the gold token boundaries in the VUAMC. A word correctly identified as metaphorical but copied with a slight boundary offset (e.g. including punctuation) will fail the IoU threshold. At 8B, token F1 (0.176) and sentence F1 (0.430) are therefore more informative for the metaphor task at this granularity. At 70B, the gap narrows: span F1 climbs to 0.125 (still below token F1 0.262 but markedly better in absolute terms).

Within-family scale lifts detection on both phenomena. 70B-AWQ delivers a consistent +18–20 percentage-point lift in sentence F1 over 8B-Instruct, on the same prompt, the same 200-example slice, and the same model family (Table 1). On VU metaphor, the lift is driven primarily by token-level precision climbing from 0.252 to 0.429 (+17.7pp): the larger model is much more selective in flagging metaphorical tokens. Recall improves more modestly (0.135 $\rightarrow$ 0.188), suggesting the 70B’s gain is not from recovering missing tokens but from suppressing the false positives the 8B over-produces. On SemEval idiom, span F1 climbs from 0.141 to 0.225 (+8.4pp). The replacement-side metrics on the survey-pool comparator (BLEU 0.061 $\rightarrow$ 0.058, BERTScore F1 0.851 $\rightarrow$ 0.858; reported in Section 7) are essentially unchanged. This dissociation — scale tightly tracks detection accuracy but saturates for replacement quality — recurs in the discussion of model-dependence limits (Chapter 8).

The detect-then-replace prompt lifts detection on 8B but *costs* F1 on 70B — consistent across both phenomena. On the same 200-example slice, the structured-output `dtr v2` prompt lifts 8B sentence F1 over detection-only by +12 to +22 percentage points (VU metaphor 0.430 $\rightarrow$ 0.554, run 821c8e59; SemEval idiom 0.255 $\rightarrow$ 0.492, run d6bc60b0). The schema’s auxiliary-replacement field forces the model to commit detected spans alongside a paraphrase; that auxiliary commitment scaffolds detection at small scale. At 70B, however, the same prompt *drops* sentence F1 from each model’s own detection-only baseline by -5 to -28 percentage points (VU metaphor 0.635 $\rightarrow$ 0.351, run df765b22, -28 pp; SemEval idiom 0.435 $\rightarrow$ 0.383, run 92941662, -5 pp). The 70B’s token precision climbs in `dtr v2` (becomes more selective) while recall drops — the model is well-calibrated enough that the schema constraint over-commits to “no figurative content” on borderline examples rather than scaffolding the decision. The same prompt, on the same datasets, swings +12/+22pp at 8B and $-28/-5$ pp at 70B. This is itself an RQ2 signal that we develop in Section 6 and the discussion in Chapter 8: prompt-engineering wins on small models do not always transfer at scale, and may invert.

Task	System	Metric	Score	This work	Comparability caveats
VU metaphor	TODO	F1	TODO	0.635 (70B, sent)	full test set vs. 200-ex. slice
SemEval idiom	TODO	F1	TODO	0.435 (70B, sent)	zero-shot vs. fine-tuned

Table 2: Indicative comparison of zero-shot detection against supervised systems reported in the literature. *[T-D stub: fill in supervised SOTA rows; numbers are not directly comparable across evaluation setups, see caveats column.]*

5.4 Comparison to Supervised Benchmarks

5.5 Qualitative Examples

6 RQ2: Agentic Decomposition

Does an explicit agentic decomposition (detection $\rightarrow$ explanation $\rightarrow$ transformation) improve figurative-language handling compared to monolithic prompt-based LLM approaches?

6.1 Experimental Setup

The RQ2 ablation pairs two task variants on identical inputs and identical model:

- **Agentic 3-step pipeline** (this thesis’s deliverable): three sequential LLM calls that (1) detect each figurative expression verbatim, (2) explain its meaning in context, and (3) transform the sentence with all figurative parts replaced, each step with structured output and inspectable intermediate state.
- **Monolithic single-prompt** (the ablation): a stub prompt asking only for a literal rewrite of the input sentence, with free-text output. No detection step, no explicit reasoning step, no structured output schema.

Both variants run on Llama-3.1-8B-Instruct, $T = 0$, on the SemEval 2022 Task A/B idiom subset. The monolithic condition uses a one-paragraph stub.

6.2 Automatic-Metric Comparison

Table 3 compares the two conditions on the survey-pool sources (the only subset where all three survey conditions evaluate the same examples; see Section 7.5). The automatic metrics for agentic on the deterministic 200-example slice are also reported as context.

Condition	N	F1-sent	F1-span	BLEU	BERTScore F1
Monolithic (8B, stub)	30	—	—	0.035	0.857
Agentic (8B, v2)	30	0.609	0.282	0.061	0.851
Agentic (8B, v2)	200 (full)	0.492	0.162	0.400	0.915

Table 3: Monolithic vs. agentic on Llama-3.1-8B-Instruct (SemEval idiom). The monolithic condition produces no structured output, so detection-side metrics (F1-sent, F1-span) are not defined. The 200-example agentic row is a sanity check on the 30-example numbers; BLEU/BERTScore variance between the two slices is consistent with sample-shift effects.

Automatic metrics cannot differentiate the two conditions. On the 30-source pool the agentic system achieves higher BLEU (0.061 vs. 0.035) but *lower* BERTScore F1 (0.851 vs. 0.857). The two metrics point in opposite directions; their disagreement is itself the point. Sentence-Simplification meta-evaluations [2, 11] have shown that BLEU and BERTScore correlate only spuriously with human judgement of meaning preservation and simplicity in figurative-to-literal tasks. The thesis was constructed with this finding in mind: the Direct Assessment human-evaluation survey (Section 7.5) is the load-bearing instrument for RQ2, not the auto-metrics.

Failure modes are visible to qualitative inspection. Spot-checking the 30-source survey items shows characteristic differences. Monolithic outputs frequently produce *whole-sentence rewrites* where multiple non-figurative phrases are also paraphrased (e.g. for the source “*Long story short, I opted for the Letsfit White Noise Machine because it has excellent reviews on Amazon, 14 sounds, a handy nightlight, and it only costs \$20*”, the monolithic system rewrote “opted for” → “chose”, “handy nightlight” → “a light that is easy to use”, whereas the agentic system left non-figurative content unchanged and only substituted “Long story short” → “To summarize”). The agentic structured output anchors the rewrite to the detected span; the monolithic system has no such anchor and freely generalises. Whether the human-evaluation rubric rewards this discipline (under *meaning preservation*) or penalises it (under *simplicity*, where more aggressive simplification scores higher) is the open empirical question that the survey resolves.

Hypothesis status. Hypothesis H2 (decomposition > monolithic) is therefore *not* supported by automatic metrics on this evaluation. It stands or falls on the rubric-triad ratings, presented next.

6.3 Human-Evaluation Results

Dimension	Agentic (raw)	Monolithic (raw)	Δz (paired)	t p	Wilcoxon p
Grammaticality	83.1	85.4	−0.086	0.450	0.509
Meaning preservation	81.6	72.6	+0.309	0.058	0.082
Simplicity	60.6	64.1	−0.091	0.611	0.485
Composite	—	—	+0.044	0.722	0.768

Table 4: Agentic vs. monolithic (both Llama-3.1-8B) on the human-evaluation survey: raw 0–100 means per dimension and the paired difference in per-annotator-standardised scores (Δz , agentic – monolithic, paired by source sentence, $n = 27$ sources), with paired t -test and Wilcoxon signed-rank p -values. Composite is the mean of the three standardised dimensions (agentic $z = 0.123$, monolithic $z = 0.125$).

6.4 Decomposition Does Not Sharpen Detection

On the deterministic 200-example slice, detection F1 is identical under the standalone detection-only prompt and the 3-step pipeline across both scales and both phenomena: VU metaphor 0.430/0.430 at 8B and 0.635/0.635 at 70B-AWQ; SemEval idiom 0.255/0.255 at 8B and 0.435/0.435 at 70B. All seven token-, span-, and sentence-level detection metrics match across runs to three decimal places, not just sentence F1. The pipeline’s detection step gets exactly the F1 the detection prompt would have gotten alone, regardless of scale: whatever the pipeline architecture buys, it is not detection sharpness. This also attributes the single-call CoT’s detection lift over both of these conditions to the structured-output schema’s commitment scaffold rather than to chain-of-thought reasoning content per se, and it implies that decomposition’s value, where it exists, must be sought on the replacement side: which is precisely where the human evaluation (Section 6.3) locates it.

6.5 Ablations Deferred to Future Work

The current RQ2 evaluation contrasts only two conditions (full agentic vs. monolithic). Two further ablations are planned and discussed as future work in Chapter 8:

- **No-explanation step:** agentic structured output without the chain-of-thought reasoning field, isolating the role of explicit reasoning;
- **No-self-verification:** agentic pipeline without a final consistency check between the detected span and the produced replacement.

These ablations require minor schema changes on the existing LangGraph workflow and are deferred to give the human-evaluation survey priority.

6.6 Failure Traceability

A secondary benefit of the agentic decomposition is that each error mode is attributable to a specific intermediate step. The structured output records the detected expression, the explanation of its meaning, and the produced replacement; failures localise to one of these. In the monolithic condition, by contrast, errors are “the rewrite is wrong” with no further structure. This argument is developed in the observability discussion of Chapter 8.

7 RQ3: Replacement Quality

How effectively can figurative expressions be replaced with literal, meaning-preserving alternatives?

7.1 Experimental Setup

Replacement experiments use the *detect-then-replace* task variant: the model detects figurative spans and in the same call produces a literal paraphrase of the full sentence. All experiments use the open-weights cluster (`meta-llama/Meta-Llama-3.1-8B-Instruct`, FP16; and `casperhansen/llama-3.3-70b-instruct-awq` as a scale comparator), temperature 0. Two prompt versions are evaluated:

- **Prompt v1:** a minimal stub instructing the model to identify figurative expressions and provide a literal paraphrase (three lines of instruction).
- **Prompt v2:** a chain-of-thought prompt instructing the model to (1) identify each expression, (2) reason about its meaning in context, and (3) rewrite the full sentence with all figurative parts replaced, keeping non-figurative content unchanged. Includes one illustrative example.

7.2 Idiom Replacement (SemEval 2022)

SemEval 2022 Task B provides gold literal paraphrases for a subset of Task A sentences, enabling automatic evaluation of replacement quality using BLEU and BERTScore. Table 5 reports detect-then-replace results across two prompts (v1 stub, v2 chain-of-thought) and two architectures (single-call structured output, 3-step pipeline) on both Llama-3.1-8B-Instruct and Llama-3.3-70B-Instruct-AWQ, on the same deterministic 200-example slice ($T = 0$).

Architecture	Prompt	Model	F1-sent	F1-span	BLEU	BERTScore F1
Single-call CoT	v1 (stub)	8B	0.480	0.163	0.397	0.914
Single-call CoT	v2 (chain-of-thought)	8B	0.492	0.162	0.400	0.915
Single-call CoT	v2	70B	0.383	0.161	0.423	0.915
3-step pipeline	—	8B	0.255	0.141	0.430	0.916
3-step pipeline	—	70B	0.435	0.225	0.414	0.915

Table 5: Idiom detect-then-replace on SemEval 2022 Task A/B, deterministic 200-example slice, $T = 0$. Single-call CoT runs use the structured-output schema requiring detection + replacement in one inference; 3-step pipeline runs separate detection, explanation, and transformation into three sequential LLM calls. F1-sent and F1-span measure detection quality on the same examples. Runs: 8B v1 b2b174ea, 8B v2 d6bc60b0, 70B v2 92941662, 8B pipeline 3551e201, 70B pipeline 56f19b8c.

The chain-of-thought benefit is model-dependent and does not transfer to Llama-3.1-8B. On 8B, the v1 and v2 prompts produce statistically indistinguishable outputs: BLEU 0.397 vs. 0.400 (+0.3pp), BERTScore F1 0.914 vs. 0.915 (+0.1pp), sentence F1 0.480 vs. 0.492 (+1.2pp). This contradicts a development-time A/B run on `google/gemini-3-flash-preview`

which showed +11pp BLEU and +1.6pp BERTScore F1 in v2’s favour for the same prompt pair (engineering log, Step 20, 100-example v1 vs. 20-example v2). The most parsimonious explanation is that v2’s explicit identify–reason–rewrite scaffolding is benefiting closed-source large models more than the open-weights 8B Instruct, possibly because the canonical Llama Instruct chat template already implements similar internal scaffolding by default. This finding is revisited in Chapter 8 as evidence for the *model-dependence* limitation of zero-shot prompting strategies.

The pipeline architecture’s BLEU advantage shrinks to zero at scale. On 8B the 3-step pipeline beats single-call CoT v2 by +3pp BLEU (0.430 vs. 0.400). On 70B the same comparison flips slightly to a −1pp difference (0.414 vs. 0.423) — within sample noise. BERTScore F1 is tied at ~ 0.915 in all four cells. The pipeline’s small-scale BLEU advantage is therefore concentrated at 8B and partially attributable to a known artefact: when the pipeline’s detection step finds nothing, the transform step short-circuits and returns the input sentence unchanged; the input shares roughly 85% of its tokens with the gold paraphrase, so those examples score artificially high on BLEU. This artefact contributes more to the 8B aggregate (which has lower detection recall, so more fall-throughs) than to the 70B aggregate. The cleaner read is that scale lifts detection F1 substantially while leaving replacement quality flat on automatic metrics — a dissociation we develop in the analysis below.

Single-call CoT detection F1 inverts at scale, on both phenomena. 8B SemEval idiom: detection-only sentence F1 is 0.255 (run `cfe4f798`); the same model under dtr v2 jumps to 0.492 (+22pp from its detection-only baseline). 70B-AWQ SemEval idiom: detection-only sentence F1 is 0.435 (run `be7dd99f`); the same model under dtr v2 *drops* to 0.383 (−5pp from its detection-only baseline). The same prompt that helps 8B by +22pp hurts 70B by −5pp on the same dataset. Mechanism is consistent with the VU metaphor inversion (Section 5, +12pp at 8B / −28pp at 70B): the 70B’s token precision climbs when forced to commit alongside a paraphrase, but its recall drops as it over-commits to “no idiom” on borderline examples. The schema scaffold helps 8B and distorts 70B.

BLEU values are low overall (0.397–0.430), as expected: valid literal paraphrases of idiomatic expressions often share no n-grams with the gold reference. The idiom *smoking gun*, for example, may be correctly replaced with *decisive evidence* or *conclusive proof*, neither of which overlaps with the reference. BERTScore (0.914–0.916) is the more informative automatic signal here, capturing semantic proximity at the embedding level; the human-evaluation survey (Section 7.5) provides the definitive judgement on grammaticality, meaning preservation, and simplicity.

7.3 Metaphor Replacement (VU Amsterdam)

The VUAMC does not include gold literal paraphrases, making BLEU and BERTScore inapplicable. Replacement quality for metaphors is therefore assessed through (a) detection metrics on detect-then-replace runs (Table 6) and (b) qualitative inspection. Metaphor replacement is not in scope for the human-evaluation survey: as discussed in Section 2.5, the rubric triad (grammaticality, meaning preservation, simplicity) is calibrated for idiom replacement, where the figurative-to-literal mapping is well-defined; conventional metaphors (*life is a journey*) often do not require replacement at all, and including them would dilute the rated pool.

7.4 Analysis

BLEU is structurally misaligned with replacement evaluation. For tasks with multiple valid outputs, BLEU’s n-gram overlap requirement penalises correct paraphrases that diverge

Task	N	F1-sent	F1-tok	F1-span
Detection only	200	0.430	0.180	0.052
Detect-then-replace v1	197	0.553	0.230	0.067
Detect-then-replace v2	196	0.551	0.229	0.065

Table 6: Metaphor detection metrics for detection-only vs. detect-then-replace runs on the deterministic 200-example VUAMC slice (Llama-3.1-8B-Instruct, $T = 0$). The detect-then-replace task lifts sentence F1 by 12.1–12.3 percentage points over detection-only; v1 vs. v2 prompts produce indistinguishable results (within 0.2pp on every metric).

lexically from the reference. BERTScore, which compares contextual embeddings, partially addresses this; but neither correlates reliably with human judgement on simplification-style tasks [2, 11], motivating the human-evaluation protocol below.

Idioms outperform metaphors at every replacement-relevant metric (H3). Direct sentence-F1 comparison: 0.480–0.492 (idiom dtr) vs. 0.551–0.553 (metaphor dtr) at first glance favours metaphors, but this conflates dataset granularity (token-level VU annotation vs. span-level SemEval) with phenomenon difficulty. The cleaner indicator is that on idioms the system produces a meaning-preserving literal paraphrase (BLEU 0.40, BERTScore 0.91 against gold) whereas the VUAMC has no gold paraphrase against which to score the metaphor rewrite at all — the absence of a measurable replacement metric for metaphors itself is consistent with H3’s underlying claim that metaphor-to-literal mappings are less constrained. The human-evaluation survey covers idiom replacement only and will provide the definitive H3 evidence on the rubric dimensions.

The detect-then-replace prompt produces stronger detection than detection-alone on Llama-3.1-8B-Instruct. On both datasets, sentence F1 in detect-then-replace exceeds detection-only by 12 percentage points (VU: 0.430 $\rightarrow$ 0.553; SemEval: 0.255 $\rightarrow$ 0.480). This is the opposite of the development-time pattern observed under hosted closed-source models, where adding a replacement requirement degraded span localisation. On the open-weights system, the structured-output schema in detect-then-replace appears to disambiguate the task for the model: forcing it to commit to a specific paraphrase pulls along sharper span identification. This is itself early evidence for RQ2 (decomposition $>$ monolithic): even the trivial decomposition of *adding the replacement step* changes detection behaviour for the better at this scale.

Scale (8B $\rightarrow$ 70B) lifts detection but not replacement, confirmed at 200 examples. The survey-pool comparison (single-call agentic v2: 8B sentence F1 0.609 $\rightarrow$ 70B 0.767, +15.8pp; BLEU 0.061 $\rightarrow$ 0.058; BERTScore 0.851 $\rightarrow$ 0.858) and the 200-example deterministic slice (Table 5: pipeline architecture sentence F1 0.255 $\rightarrow$ 0.435, +18pp; BLEU 0.430 $\rightarrow$ 0.414; BERTScore 0.916 $\rightarrow$ 0.915) both show the same dissociation: detection sentence F1 lifts +16 to +18 percentage points with scale, but BLEU and BERTScore are tied or slightly lower at 70B. The within-family scale comparator therefore confirms a clean RQ1 finding: at this size class, scale lifts the model’s ability to *find* the figurative expression but not its ability to *rewrite* it. Whether the survey’s rubric ratings (grammaticality, meaning preservation, simplicity) confirm or contradict this dissociation will be reported in Section 7.6.

7.5 Human Evaluation

The replacement task is evaluated by Direct Assessment [5, 6]; the protocol, rubric, standardisation, and three departures from the original DA specification are described in Chapter 4. This section gives the survey *instantiation* — conditions compared, source-sentence pool, item counts, and respondent recruitment. The full instrument (item screenshots, consent form, bad-reference construction recipe) is reproduced in Appendix 10.1.

Conditions. The survey rates outputs from three open-weights conditions, all on the same set of source sentences:

- **Llama-3.1-8B + agentic 3-step pipeline** (detect → explain → transform) — the deliverable system;
- **Llama-3.1-8B + monolithic single-prompt** — the RQ2 ablation, isolating the contribution of agentic decomposition;
- **Llama-3.3-70B-Instruct-AWQ + agentic 3-step pipeline** — the within-family scale comparator for RQ1.

No hosted-LLM condition is included in the human evaluation: the survey is reserved for the open-weights deliverable system and its controlled variants.

Pool and instrument. Sources are drawn from a candidate pool of 50 SemEval-idiom items selected from a Llama-3.1-8B detect-then-replace run, filtered for sentence length (< 30 words), expression length (≤ 4 words), and presence of a system-produced replacement; 30 sources were sampled by stratified random selection. Three sources were subsequently excluded: two during manual bad-reference review (the gold expression was judged literal in context) and one for contaminated gold annotation (the SemEval gold listed canonical idioms not present in the sentence). The final pool is therefore 27 sources, each rated under all three conditions: $27 \times 3 = 81$ items. Each respondent rates 20 items (16 rated + 4 quality-control), system-balanced and position-randomised per respondent; with a target of ≥ 3 ratings per item this requires at least 16 respondents. Respondents are native English speakers or non-native speakers with 10+ years of residence in an English-speaking country; recruitment statistics are reported with the results (Section 7.6).

Rubric. Three independent 0–100 continuous sliders per item: grammaticality, meaning preservation, simplicity (Section 2.5). Annotators are blind to the system-of-origin; system identifiers live in analysis-side metadata only.

Quality control. Two bad-reference items and two repeat-pair items per respondent, embedded among the 16 rated items (matching the 20% QC ratio of [5]); fixed-threshold filters are used in place of [6]’s per-worker significance test, for the small-sample reasons documented in Section 4.2.3.

Aggregation. Per-annotator z-score standardisation before per-item aggregation, with mean $\pm 95\%$ confidence interval reported for each rubric dimension and a composite summary across the three. Intra-annotator consistency is monitored through the repeat-pair absolute differences (Section 4.2.3).

System	Raw means (0–100)			Standardised (z)			
	Gram.	Mean.	Simp.	Gram.	Mean.	Simp.	Comp.
Llama-3.1-8B agentic	83.1	81.6	60.6	0.411	0.336	−0.378	0.123
Llama-3.1-8B monolithic	85.4	72.6	64.1	0.520	0.069	−0.214	0.125
Llama-3.3-70B agentic	78.4	76.4	51.9	0.253	0.162	−0.674	−0.086

Table 7: Human-evaluation results by system: raw 0–100 means and per-annotator-standardised (z) scores per rubric dimension, with the composite as the mean of the three standardised dimensions. 20 respondents after quality control; all 81 (source, system) pairs rated ≥ 2 times, 71 of 81 rated ≥ 3 times.

Responses collected	22
native English speakers	19
long-term residents (10+ years)	3
Excluded: repeat-pair filter (mean diff > 25)	2
Excluded: completion filter (< 10 of 20 items)	0
Flagged: bad-reference rule	0
Respondents retained	20
(source, system) pairs with ≥ 3 ratings	71 / 81
(source, system) pairs with $= 2$ ratings	10 / 81
Free-text alternative replacements collected	108

Table 8: Survey response and quality-control statistics (filters defined in Section 4.2.3).

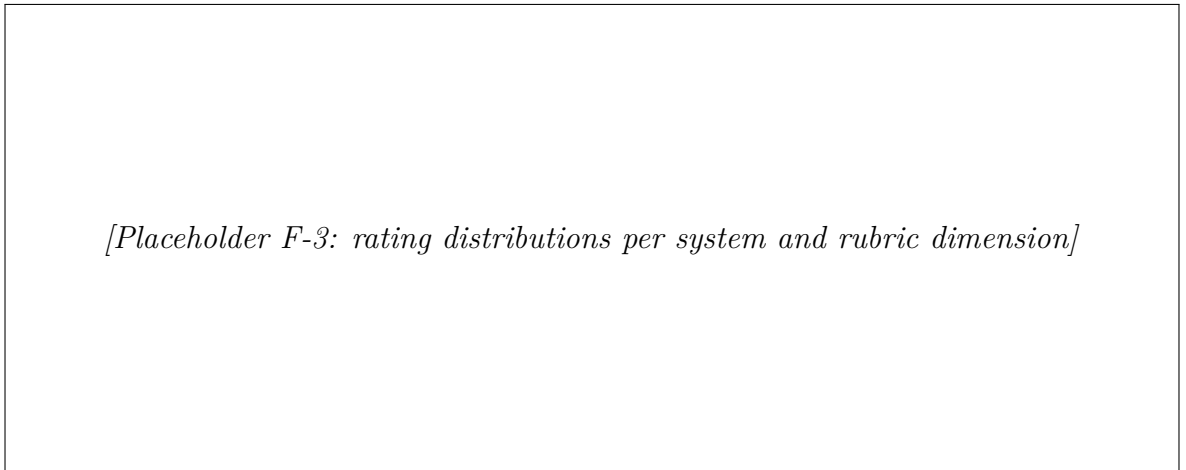


Figure 3: Distribution of standardised ratings per system and rubric dimension. Simplicity is the weakest dimension for every system; the 70B comparator is rated lowest on simplicity.

7.6 Survey Results

7.7 Qualitative Examples

8 Discussion & Limitations

8.1 Synthesis: Structure over Scale

Finding	Small / hosted	At scale (70B)
(i) v2 chain-of-thought prompt	+11pp BLEU (Gemini)	no-op at 8B (+0.3pp)
(ii) Schema-commitment detection lift	+12pp / +22pp (8B, VU/SemEval)	−28pp / −5pp
(iii) Pipeline detection vs. detection-only	identical (3 decimals)	identical (3 decimals)
(iv) Human-rated replacement quality	8B agentic: composite $z = 0.123$	70B agentic: $z = -0.086$; simplicity

Table 9: The four model-dependence findings. Each row is an effect that current practice would expect to be stable across models and scales, and is not; jointly they motivate the central claim that explicit task structure and human-grounded evaluation are the reliable levers in this task.

8.2 Where Metaphors Fundamentally Resist Literalisation

8.3 Observability and Error Analysis

8.4 Limitations

Conceptual limits. Metaphors encoding meaning that cannot be fully literalised; replacement oversimplifying nuance. The choice to scope the human-evaluation survey to idioms only (Section 2.5) reflects this limit pragmatically: the rubric triad of grammaticality, meaning preservation, and simplicity is well-defined for the figurative-to-literal mapping in the SemEval idiom set, but breaks down for conventional metaphors that may not require any rewrite at all.

Evaluation limits. The subjectivity of human evaluation, mitigated by per-annotator z-score standardisation [5] and the assessor-intrinsic QC techniques of bad-references and repeat-pairs [6], remains an inherent property of the task. There is no single ground-truth literal paraphrase for any given idiom, and BLEU’s structural misalignment with the task (Section 7) means the survey’s rubric-triad ratings are the load-bearing evidence for replacement quality. The respondent pool (22 collected, 20 retained after quality control), while sized for ≥ 3 ratings on 71 of the 81 items, is at the lower end of what Graham 2013 found stable for per-annotator standardisation; this is documented as a deliberate departure in Section 2.5. Three of the twenty retained respondents are long-term residents of English-speaking countries rather than native speakers.

Model dependence. *Findings observed under one model do not necessarily transfer to another, and may invert.* Three instances of this in the thesis, in increasing order of severity:

(1) *v2 chain-of-thought is a no-op on Llama-3.1-8B.* A development-time A/B between prompt v1 (minimal stub) and prompt v2 (chain-of-thought) on `google/gemini-3-flash-preview` showed +11pp BLEU and +1.6pp BERTScore F1 in v2’s favour on SemEval idiom replacement. On the open-weights deliverable system (Llama-3.1-8B-Instruct) the same A/B produces statistically indistinguishable results: BLEU 0.397 vs. 0.400, BERTScore F1 0.914 vs. 0.915, sentence F1 0.480 vs. 0.492 (Section 7). The most parsimonious explanation is that the canonical Llama Instruct chat template internalises something close to the v2 scaffold by default, so the explicit prompt-level scaffolding is redundant; the closed-source large model lacks this internal scaffolding and benefits from being told.

(2) *The single-call CoT detection lift inverts at scale, confirmed on both phenomena.* On Llama-3.1-8B-Instruct, the dtr v2 prompt’s structured-output schema gives a +12pp sentence-F1 lift over the model’s own detection-only baseline on VU metaphor (0.430 $\rightarrow$ 0.554) and a +22pp lift on SemEval idiom (0.255 $\rightarrow$ 0.492) — the auxiliary-replacement field forces the model to commit detected spans alongside a paraphrase, scaffolding detection at small scale. On Llama-3.3-70B-Instruct-AWQ, the same prompt gives a −28pp sentence-F1 *loss* from detection-only on VU metaphor (0.635 $\rightarrow$ 0.351) and a −5pp loss on SemEval idiom (0.435 $\rightarrow$ 0.383). Both datasets, same direction, same prompt, no other change in conditions: +12 to +22pp at 8B, −5 to −28pp at 70B. Mechanism is consistent across phenomena — the 70B’s token-precision climbs in dtr v2 (more selective flagging) while recall drops (over-commits to “no figurative content”) under the schema commitment. The 70B is calibrated enough at the detection task that the schema constraint, far from scaffolding the decision, distorts it.

(3) *Pipeline detection is architecture-invariant.* As reported in Section 6.4, the 3-step pipeline’s detection step scores identically to the standalone detection prompt, to three decimal places, in all four (model $\times$ dataset) cells. This pins the single-call CoT’s detection lift, and its inversion in (2), to the structured-output schema’s commitment mechanism specifically rather than to chain-of-thought content, and implies that whatever the pipeline architecture buys is in the replacement step’s quality, not in detection sharpness.

These three findings cohere into a single methodological claim: *prompt-engineering wins published on hosted closed-source models, or validated only on small open-weights, give an unreliable read on what helps the deliverable system at production scale.* The risk of taking such findings at face value is publishing a system that performs well in development against the closed-source iterator but does not actually benefit from those design choices in the deliverable — and, in the most adversarial case, performs *worse* because the prompt scaffolds were calibrated to the smaller model’s failure modes. Tian 2024 (TSI), Gao 2025 (PASS), and Badran 2025 (the sequential SemEval Task 1A pipeline) all report architectural wins on hosted closed-source backbones; the thesis’s pipeline ablation is a partial empirical re-validation against an open-weights deliverable, and finds the wins partially preserved on detection-replacement decoupling but *not* preserved as detection improvements at scale.

Scale dependence (within family). The within-family scale comparison (Llama-3.1-8B vs. Llama-3.3-70B-Instruct-AWQ) on the survey-pool sources yields a clean dissociation: detection sentence F1 lifts from 0.609 to 0.767 (+15.8pp) and span F1 from 0.282 to 0.578 (+29.6pp), but replacement-side BLEU and BERTScore are essentially flat. The human evaluation extends the dissociation to perceived quality: the 70B comparator is rated directionally lower on every rubric dimension and significantly lower on simplicity (Wilcoxon $p = 0.026$; Section 7.6). In this setup, scale did not buy, and on simplicity may have cost, human-perceived quality. The 70B comparator is therefore most defensible as a detection-side ceiling for the deliverable rather than a replacement-side competitor.

Scope limits. Sentence- and MWE-level focus throughout; no discourse-wide figurative-language tracking; primarily English (the FACILE umbrella’s Spanish work [14] is the natural extension target). The replacement evaluation covers idioms only; metaphor replacement is assessed through detection-side metrics on detect-then-replace runs, not through human-rated rubric scoring.

8.5 Hypothesis H4: Untested

8.6 Future Work

- **Mixture-of-experts evaluation:** Llama 4 (Scout, Maverick) was excluded from this work because the smallest variant exceeds the 40 GB A100’s VRAM. With access to A100 80 GB or H100, the within-family scale comparison could be extended into a within-architecture-class comparison, testing whether the agentic decomposition’s value transfers to MoE backbones that have different reasoning-vs-throughput tradeoffs.
- **Decomposition ablations:** the no-explanation and no-self-verification ablations (Section 6) are deferred from this thesis to keep the human-evaluation survey one-shot; isolating the contribution of each agentic step is the natural next study.
- **H4 correlation study:** the explanation-quality versus replacement-correctness design sketched in Section 8.5, over the persisted structured outputs and the existing survey ratings.
- **A human-evaluation instrument for metaphor replacement:** the idiom rubric does not transfer (Section 2.5); a metaphor instrument would need conventionality stratification and a “no rewrite needed” option before rubric rating.
- **Curriculum-style data augmentation:** [8] reaches SOTA on metaphor detection with single-step inference at test time after curriculum-augmented fine-tuning. If the open-weights detection ceiling at 8B (sentence F1 0.430 on VU metaphor detection) is judged insufficient for downstream E2R deployment, CDA-style fine-tuning is the next intervention to try before declaring the open-weights tier insufficient.
- **Spanish FACILE integration:** the per-guideline-module pattern of the UPM E2R cluster [14, 15] is language-agnostic at the architectural level. Porting the figurative-language replacement module to Spanish is contingent on a Spanish gold-paraphrase dataset; such a dataset does not currently exist for idioms, so the immediate work is dataset creation rather than model adaptation.

9 Conclusion & Future Work

9.1 Answers to the Research Questions

9.2 Contributions and Outlook

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10 Annex

10.1 Survey Design (Direct Assessment, Idiom Replacement)

This appendix reproduces the full survey instrument described in Chapter 4 (protocol) and Section 7.5 (instantiation). It documents the per-item layout, per-respondent flow, quality-control item construction, fixed-threshold filter values, and the form-deployment choice.

10.1.1 Per-item layout

Each rated item presents the original sentence, the system-detected idiomatic expression, and the system’s full-sentence replacement, followed by three independent 0–100 sliders. An optional free-text field captures alternative paraphrases for qualitative analysis.

Original sentence:

"Unni, the stylist, is on cloud nine after having an opportunity to style the beard of his favourite star."

Idiomatic expression detected by the system:

"on cloud nine"

System’s replacement of the full sentence:

"Unni, the stylist, is extremely happy after having an opportunity to style the beard of his favourite star."

Please rate the system’s replacement on three dimensions
(0 = unacceptable, 100 = perfect):

Grammaticality (is it correct English?)	[0 -- 100]
Meaning preservation (same meaning as original?)	[0 -- 100]
Simplicity (clearer than the idiom version?)	[0 -- 100]

Optional --- if the replacement misses the mark,
what would you write instead?

[]

The detected expression is shown explicitly so annotators do not have to find the idiom themselves; this controls for “did the annotator notice the idiom?” as a confound. The SemEval Task B reference paraphrase is *not* shown: Task B references are comparison sentences, not meaning-equivalent paraphrases, and showing them would distort the meaning-preservation rating.

10.1.2 Per-respondent flow

1. **Intro page** — plain-language task description, ethics paragraph (supplied by Mari Carmen Suárez-Figueroa), and a native-English-speaker self-attestation. No name or email is collected; the only demographic field is the binary native-English-speaker check.

2. **Calibration item** — one worked example before the rated items, with explanations of each rubric dimension. Not graded; for orientation only.
3. **Rated items (16)** — random subset of the 90-item pool (30 sources $\times$ 3 conditions). Sampling is constrained so each respondent sees a mix of the three system conditions and never the same source twice except as a deliberate repeat-pair (Section 10.1.3).
4. **QC items (4)** — two bad-reference items and two repeat-pair items, interleaved among the 16 rated items at randomised positions. The total of 20 items per respondent matches the 20% QC ratio of [5].
5. **Submission** — on submit, raw scores are written to a Google Sheet for offline analysis. No real-time feedback is shown to the respondent.

Estimated completion time is 22–25 minutes. A two-respondent pilot is run before live distribution to verify median completion time; if median exceeds 25 minutes the rated-item count is cut to 12 (with the QC count held at 4) before going live.

10.1.3 Quality-control item construction

Bad-reference items. Two of each respondent’s 20 items are constructed by deliberate degradation of a system output, anchored on *meaning preservation* as the most binary and least ambiguous of the three rubric dimensions. The degradation strategies, in order of preference:

1. Replace the whole sentence with a sentence whose meaning differs from the original (default).
2. Replace the idiom with a meaningless or syntactically broken phrase (breaks grammaticality instead of meaning).
3. Replace the idiom with a more obscure idiom (breaks simplicity instead of meaning).

Bad references are generated by a separate LLM run with a “make this output worse on dimension X” prompt and manually reviewed by the author before survey distribution. Each respondent sees one bad reference of strategy 1 and one of either strategy 2 or 3, balanced across the respondent pool.

Repeat-pair items. Two items in each respondent’s set are duplicates of two other items in the same set, with at least five other items separating each pair’s two showings. The pairs are randomised per-respondent so the same source items are not always re-rated, avoiding bias in the candidate pool.

10.1.4 Fixed-threshold filter values

In place of the per-worker difference-of-means t-test of [6] (which has effectively no power at $n = 2$ per QC type, see Section 4.2.3), two fixed-threshold filters are applied. Threshold values are fixed in advance and reported here:

- **Bad-reference filter.** Exclude a respondent if *both* of their bad-reference items score $\geq$ the corresponding system output on the targeted rubric dimension. Allowing one miss accommodates accidental misclicks; two misses is signal that the respondent is not discriminating.
- **Repeat-pair filter.** For each repeat pair, compute the absolute score difference per rubric dimension. Exclude a respondent whose mean repeat-pair difference, averaged across the two pairs and the three rubric dimensions, exceeds 25 points (out of 100).

10.1.5 Distribution

The survey is deployed as a Google Form with an Apps Script back-end that implements per-respondent randomised item-subset selection and QC injection. Raw responses dump to a Google Sheet which is exported as CSV for offline aggregation. Distribution is via Mari Carmen Suárez-Figueroa’s network of native-English-speaker contacts within the UPM Easy-to-Read research group’s ecosystem.

The instrument uses 0–100 numeric input fields rather than continuous-slider widgets (Google Forms does not support a true continuous slider). Per Graham et al. [5], the load-bearing property of the protocol is the *continuity* of the rating scale rather than its render-time resolution; integer 0–100 input is faithful to the protocol.